Motor Control Board

2026-09-11

Variant: DRAFT

Rev + (Unreleased)

TOP VIEW

kibot_Mmage_png_3d_viewer_angled_top

BOTTOM VIEW

kibot_Mmage_png_3d_viewer_angled_bottom

NOTES

Block Diagram

File: Block Diagram.kicad_sch

Revision History

File: Revision History.kicad_sch

Project Architecture

File: Project Architecture.kicad_sch

Power - Sequencing

File: Power - Sequencing.kicad_sch

Comment

Not fitted components are marked as ✗

DRAFT - Very early stage of schematic, ignore details.

PRELIMINARY - Close to final schematic.

CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.

RELEASED - A board with this schematic has been sent to production.

Date: 11-Sep-2026

DESIGN CONSIDERATIONS

DESIGN NOTE: Example text for informational design notes.

DESIGN NOTE: Example text for debug notes.

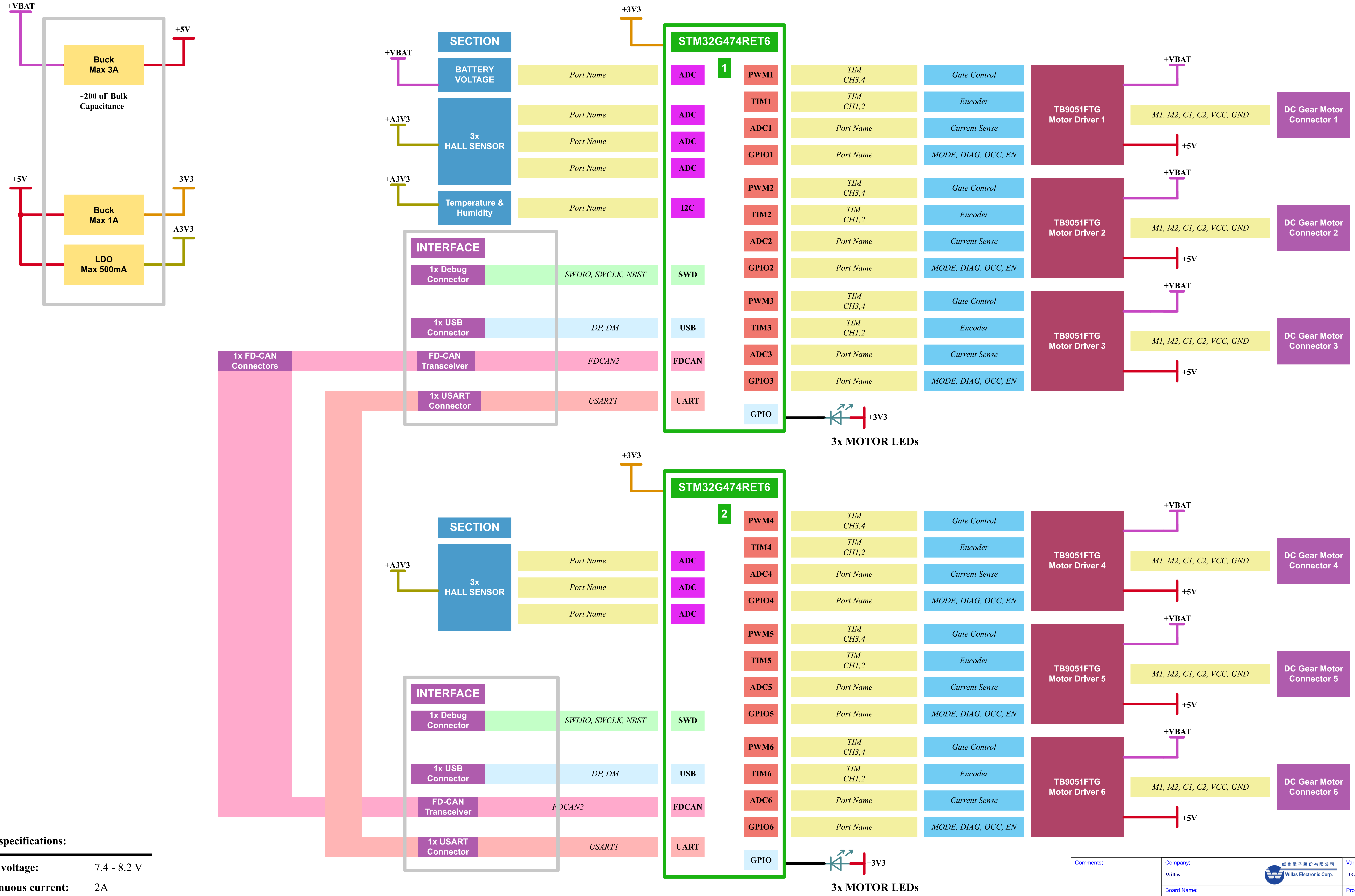
DESIGN NOTE: Example text for cautionary design notes.

DESIGN NOTE: Example text for critical design notes.

LAYOUT NOTE: Example text for critical layout guidelines.

Comments:	Company: Willas (威倫電子股份有限公司 Willas Electronic Corp.)		Variant: DRAFT	Git Hash: 309c1de
	Board Name: **Motor Control Board**		Project Name: **Master Hand**	
Sheet Title:	File Name: Master_Hand_Kicad.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /		Reviewer:	Size: A3	Sheet: 1 of 23

[2] Block Diagram

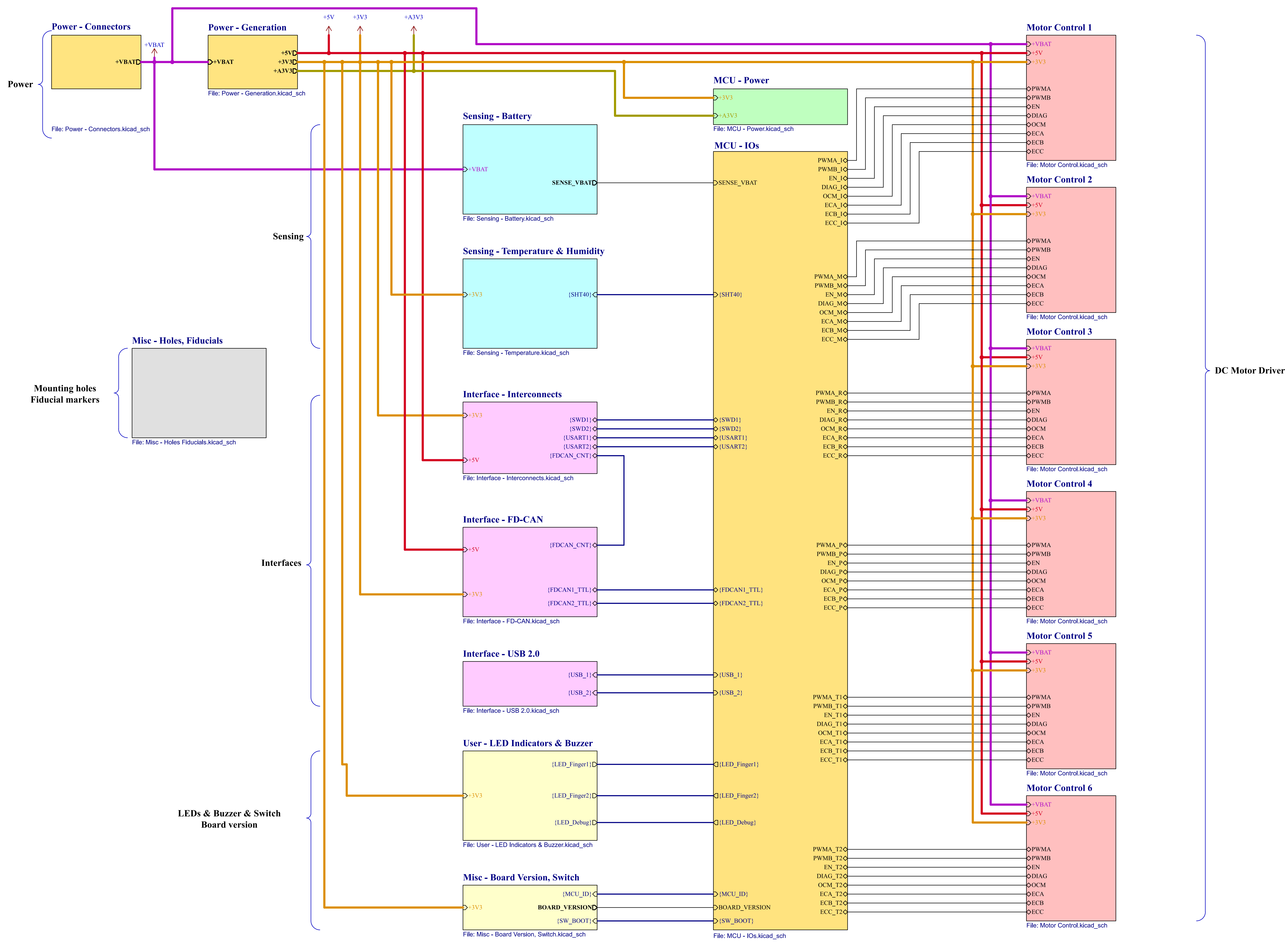


Target specifications:

Input voltage:	7.4 - 8.2 V
Continuous current:	2A
Peak current:	4A
Switching frequency:	600kHZ - 1Mhz

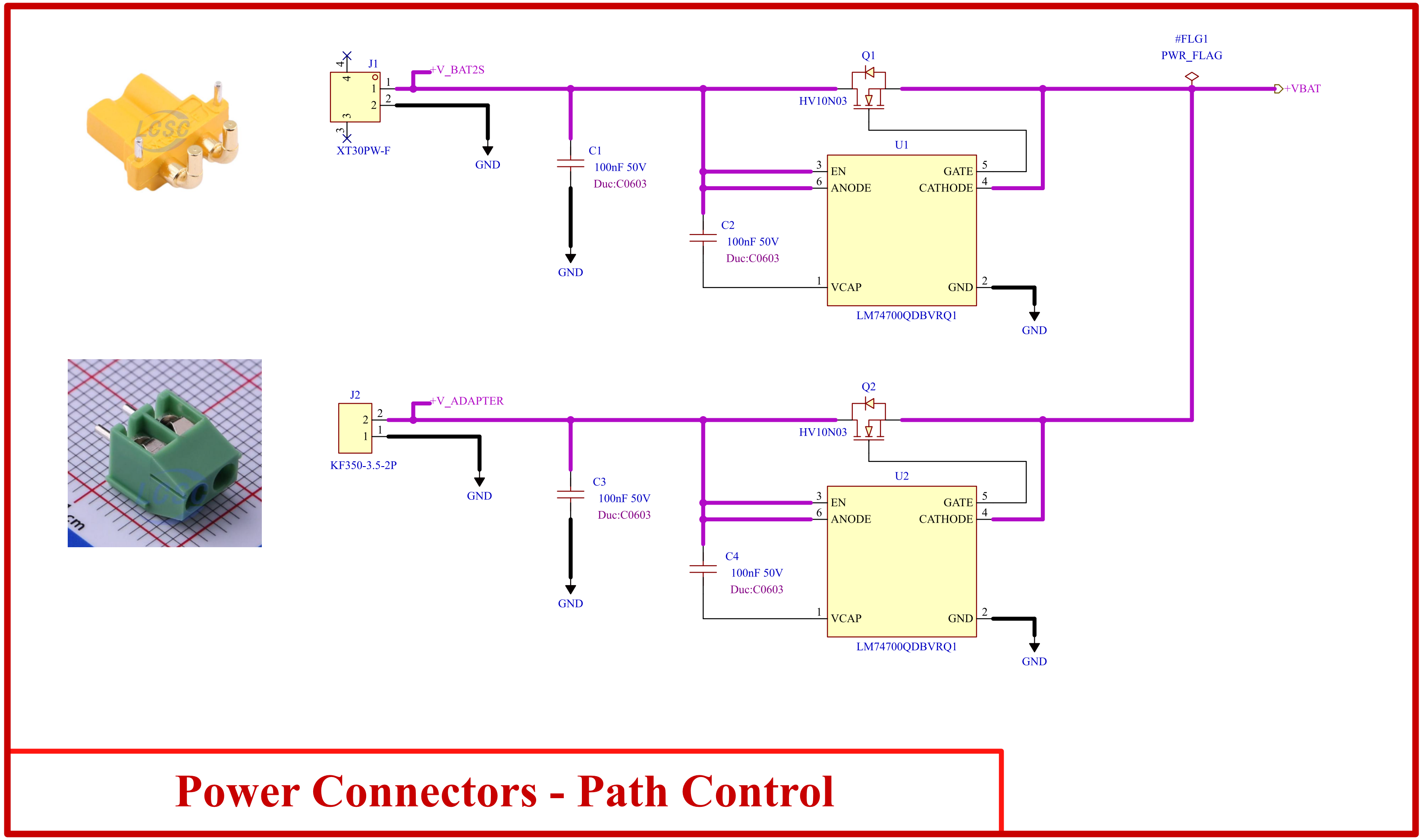
Comments:	Company: Willas		Variant: DRAFT	Git Hash: 309c1de
	Board Name: Motor Control Board		Project Name: Master Hand	
Sheet Title:	File Name: Block Diagram.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Block Diagram/		Reviewer:	Size: A2	Sheet: 2 of 23

[3] Project Architecture



Comments:	Company: Willas (威倫電子股份有限公司 Willas Electronic Corp.)	Variant: DRAFT	Git Hash: 309c1de
	Board Name: **Motor Control Board**	Project Name: **Master Hand**	
Sheet Title:	File Name: Project Architecture.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/	Reviewer:	Size: User	Sheet: 3 of 23

[4] Power - Connectors



Comments:

Company: Willas

威倫電子股份有限公司 Willas Electronic Corp.

Variant: DRAFT

Git Hash: 309c1de

Board Name: **Motor Control Board**

Project Name: **Master Hand**

Sheet Title:

File Name: Power - Connectors.kicad_sch

Designer: Damon

Date: 2026-06-27

Revision: + (Unreleased)

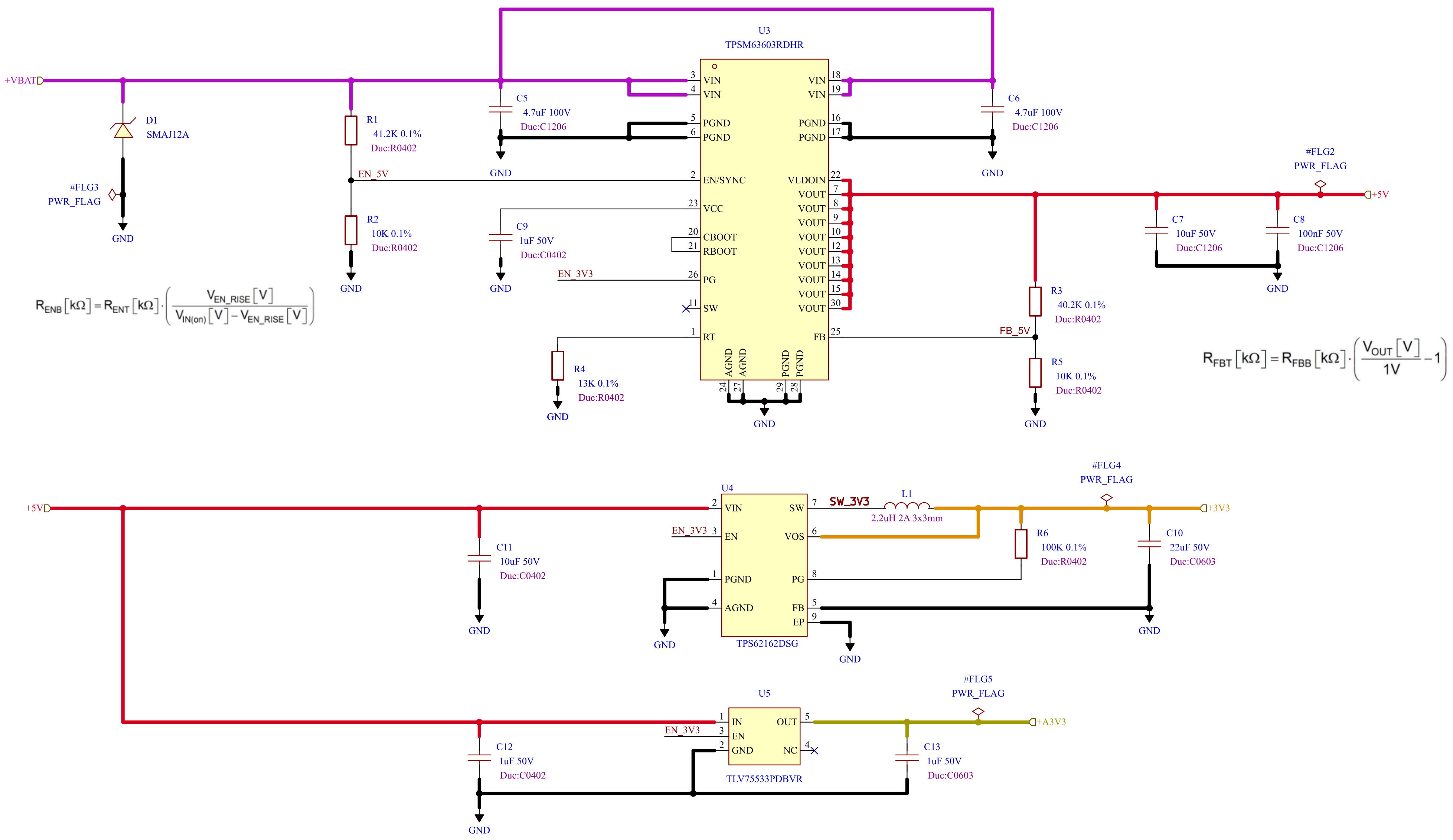
Sheet Path: /Project Architecture/Power - Connectors/

Reviewer:

Size: A4

Sheet: 4 of 23

[5] Power - Generation



$$R_{ENB}[k\Omega] = R_{ENT}[k\Omega] \cdot \left(\frac{V_{EN\_RISE}[V]}{V_{IN(on)}[V] - V_{EN\_RISE}[V]} \right)$$

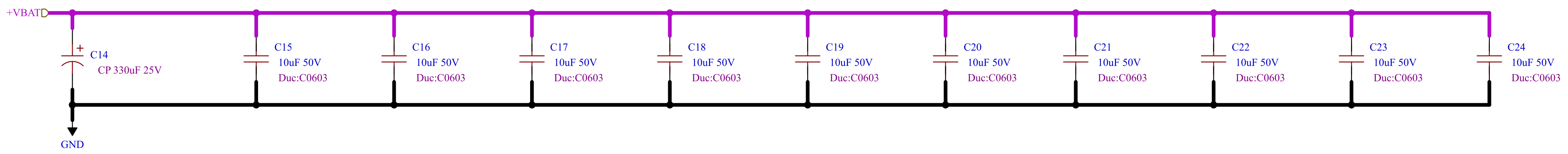
$$R_{FBT}[k\Omega] = R_{FBB}[k\Omega] \cdot \left(\frac{V_{OUT}[V]}{1V} - 1 \right)$$

DESIGN NOTE:
Pi filter with inductor would give better EMC performance but has higher bulk capacitance requirements. Ferrite is a good compromise.

LAYOUT NOTE:
Route feedback traces away from switching node.

LAYOUT NOTE:
Minimize switching node loop areas.

Voltage Regulators

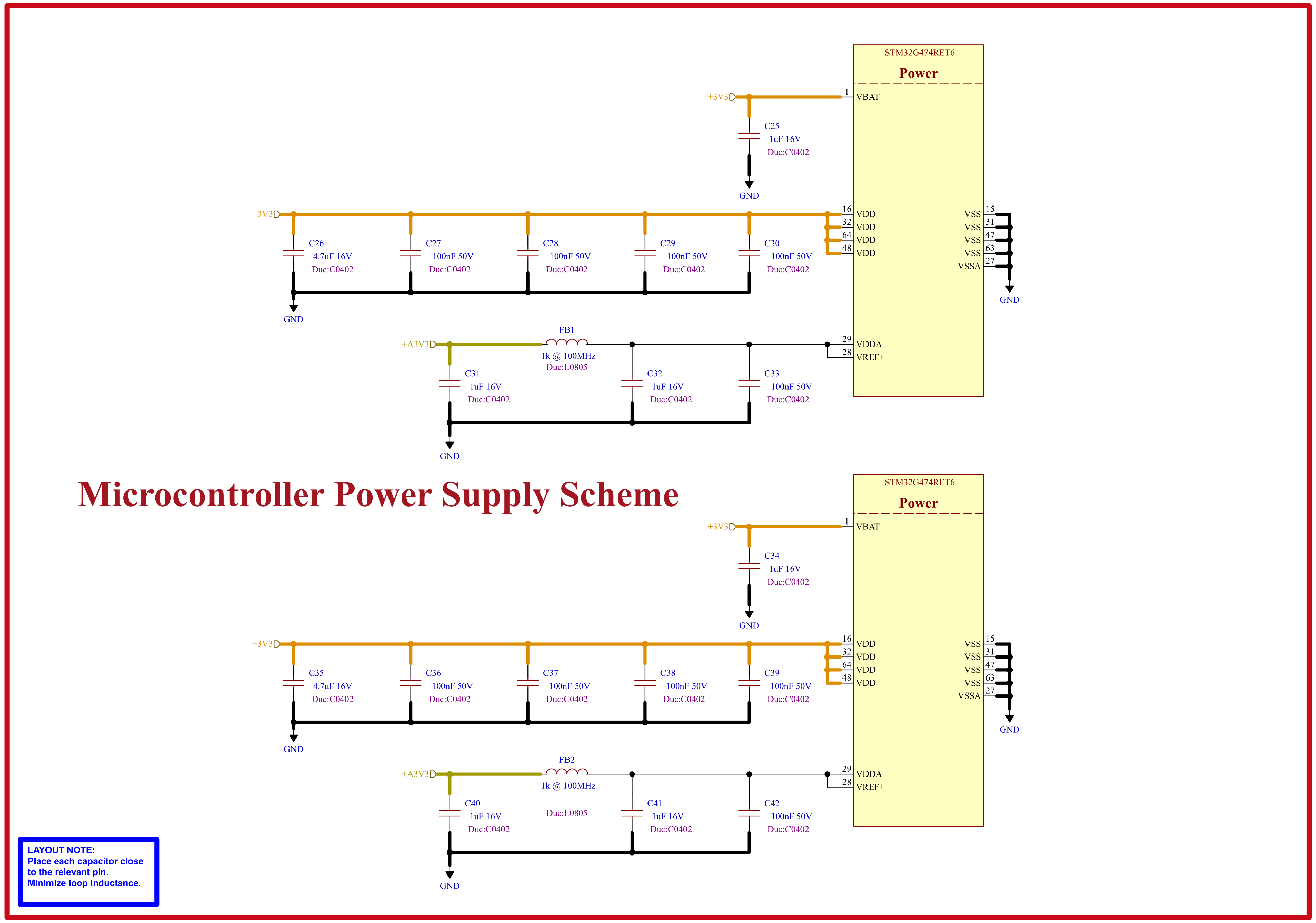


DC BULK

DC-Link Bulk Capacitors. 330uF + 10 x 10 uF = 430uF (35V)

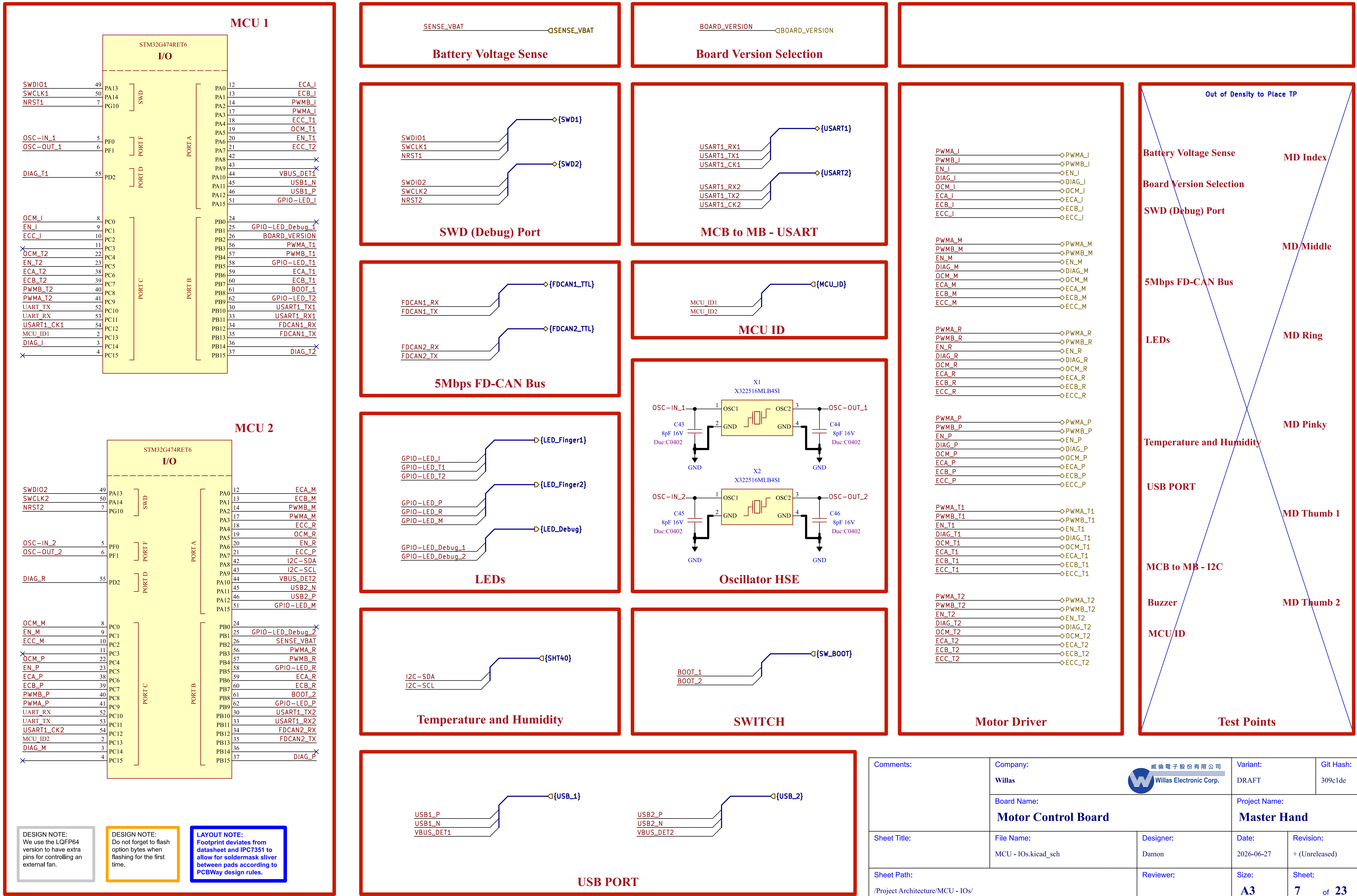
Test Points

Comments:	Company: Willas (Willas Electronic Corp.)		Variant: DRAFT	Git Hash: 309c1de
	Board Name: **Motor Control Board**		Project Name: **Master Hand**	
Sheet Title:	File Name: Power - Generation.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Power - Generation/		Reviewer:	Size: A3	Sheet: 5 of 23



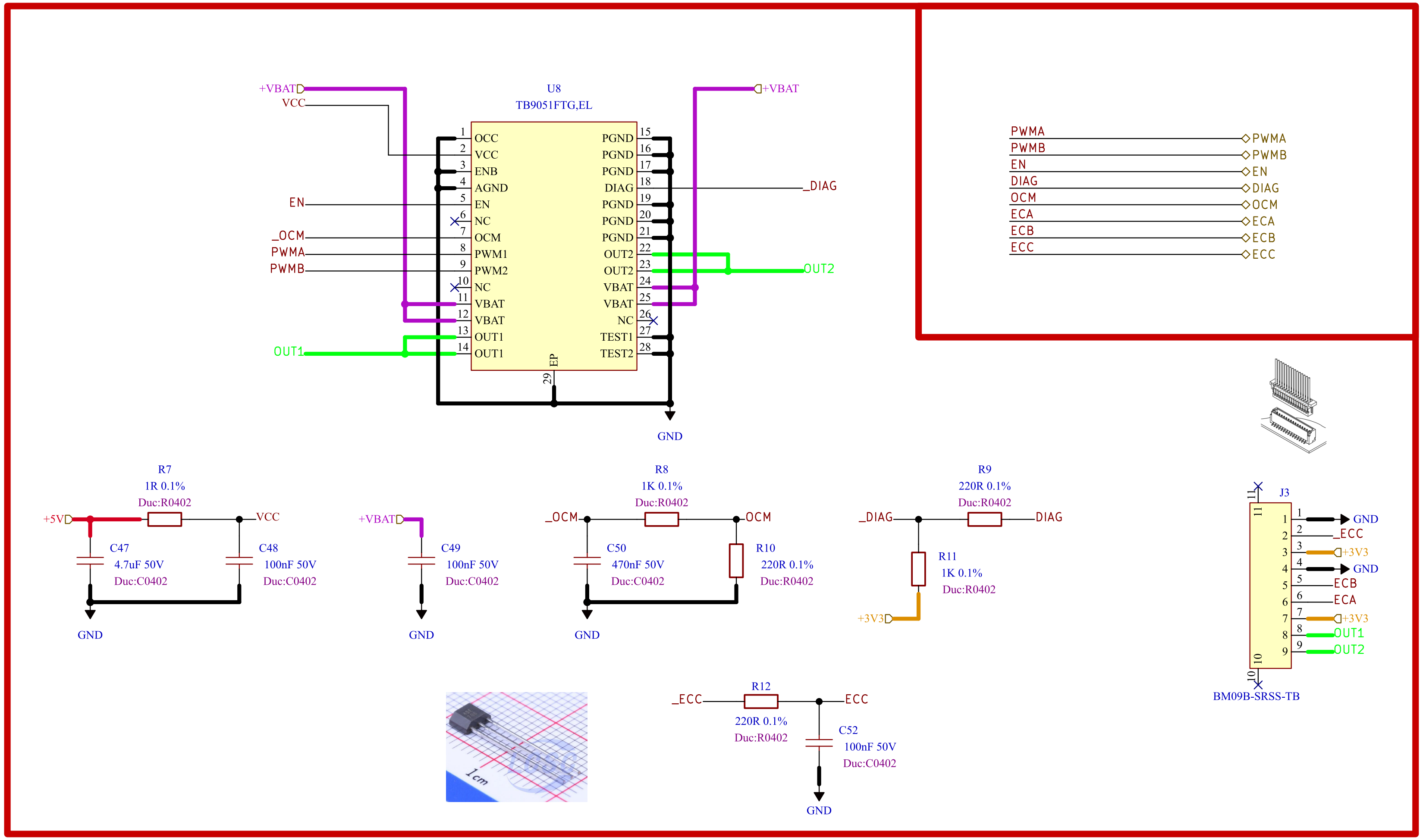
Comments:	Company: Willas	威倫電子股份有限公司 Willas Electronic Corp.	Variant: DRAFT	Git Hash: 309c1de
	Board Name: **Motor Control Board**		Project Name: **Master Hand**	
Sheet Title:	File Name: MCU - Power.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/MCU - Power/		Reviewer:	Size: A3	Sheet: 6 of 23

[7] MCU - IOs



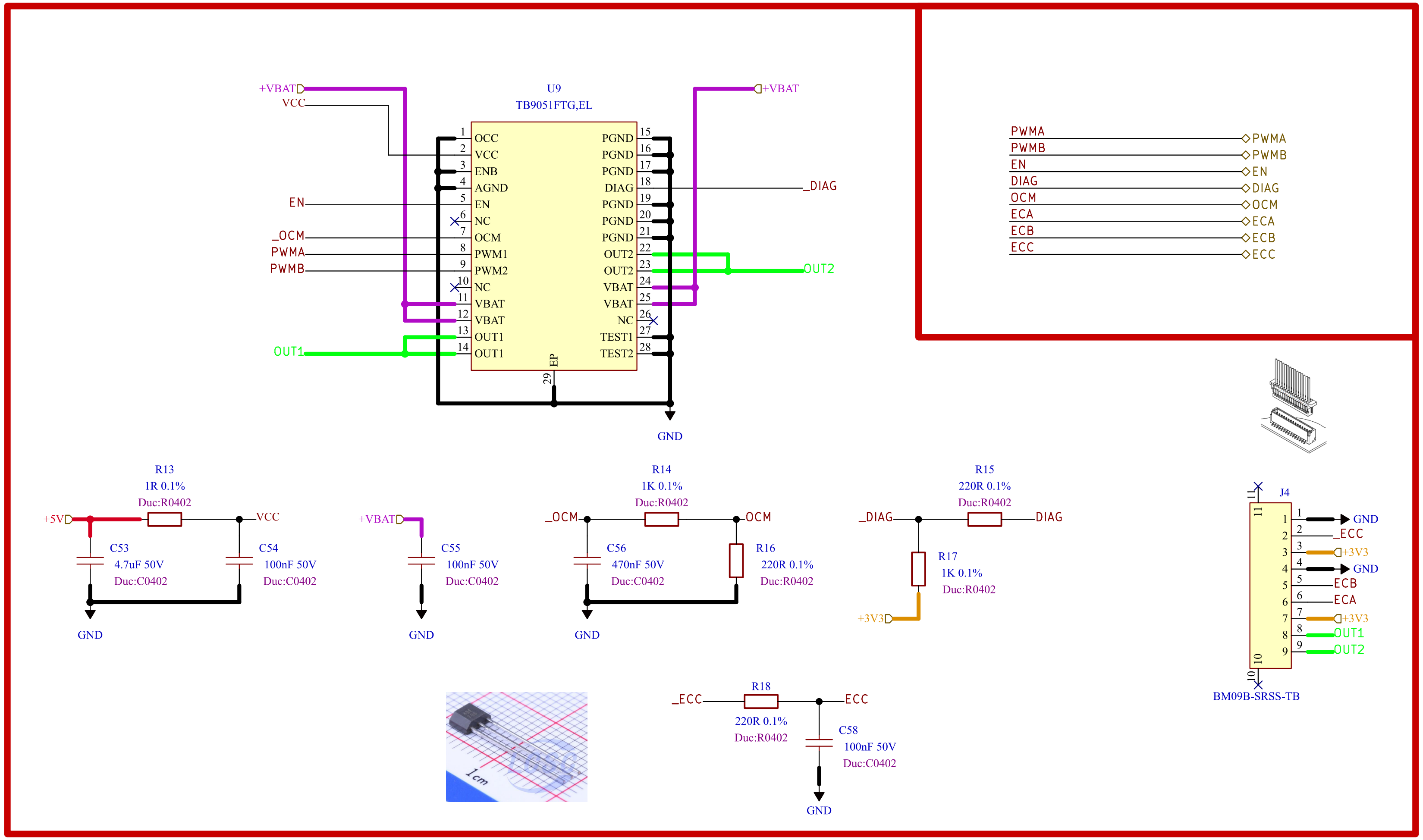
Comments:	Company: Willas 威倫電子股份有限公司 Willas Electronic Corp.		Variant: DRAFT	Git Hash: 309c1de
	Board Name: **Motor Control Board**		Project Name: **Master Hand**	
Sheet Title:	File Name: MCU - IOs.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/MCU - IOs/		Reviewer:	Size: A3	Sheet: 7 of 23

[8] Motor Control 1



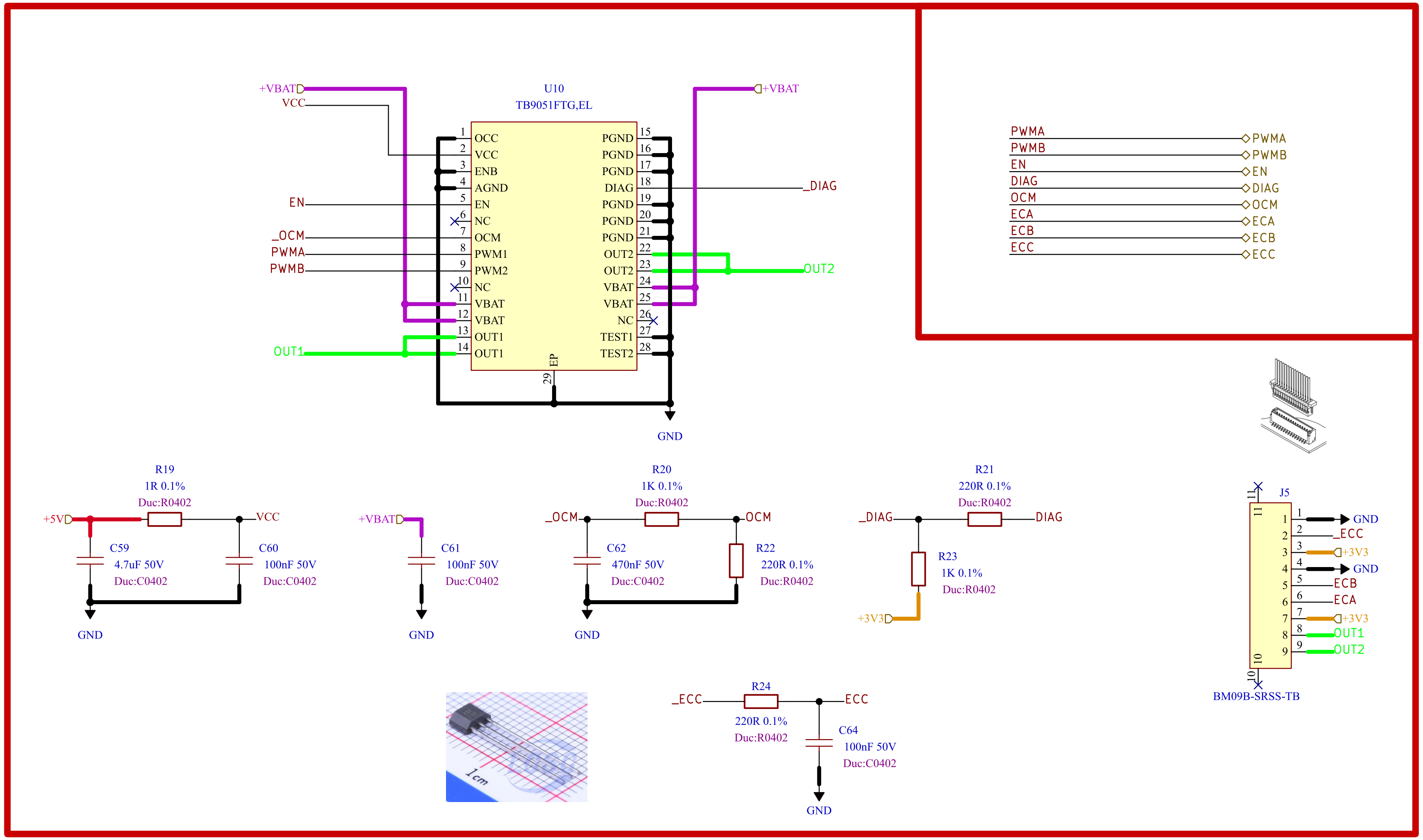
Comments:	Company: Willas (威倫電子股份有限公司 Willas Electronic Corp.)		Variant: DRAFT	Git Hash: 309c1de
	Board Name: **Motor Control Board**		Project Name: **Master Hand**	
Sheet Title:	File Name: Motor Control.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Motor Control 1/		Reviewer:	Size: A4	Sheet: 8 of 23

[9] Motor Control 2



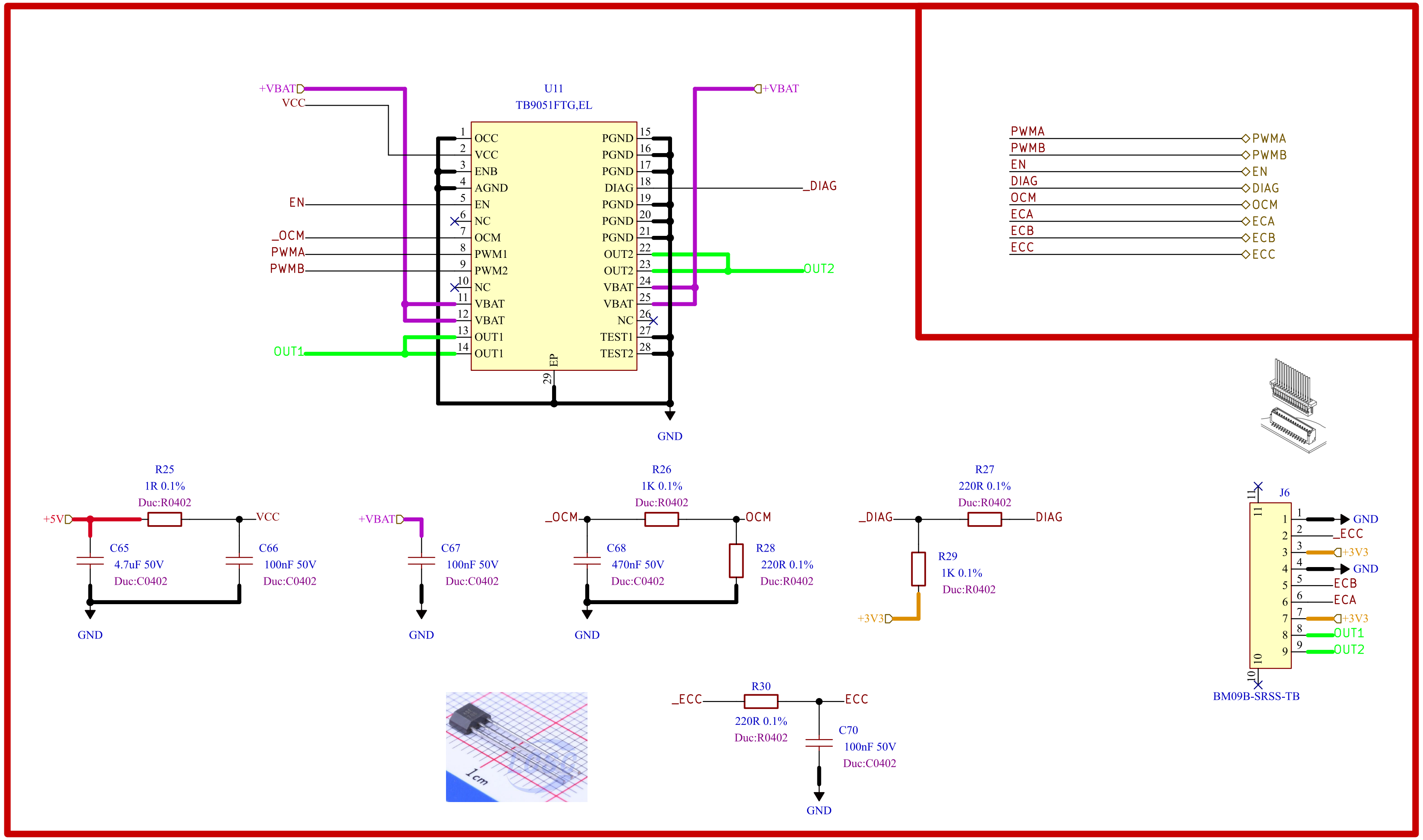
Comments:	Company: Willas (威倫電子股份有限公司 Willas Electronic Corp.)		Variant: DRAFT	Git Hash: 309c1de
	Board Name: **Motor Control Board**		Project Name: **Master Hand**	
Sheet Title:	File Name: Motor Control.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Motor Control 2/		Reviewer:	Size: A4	Sheet: 9 of 23

[10] Motor Control 3



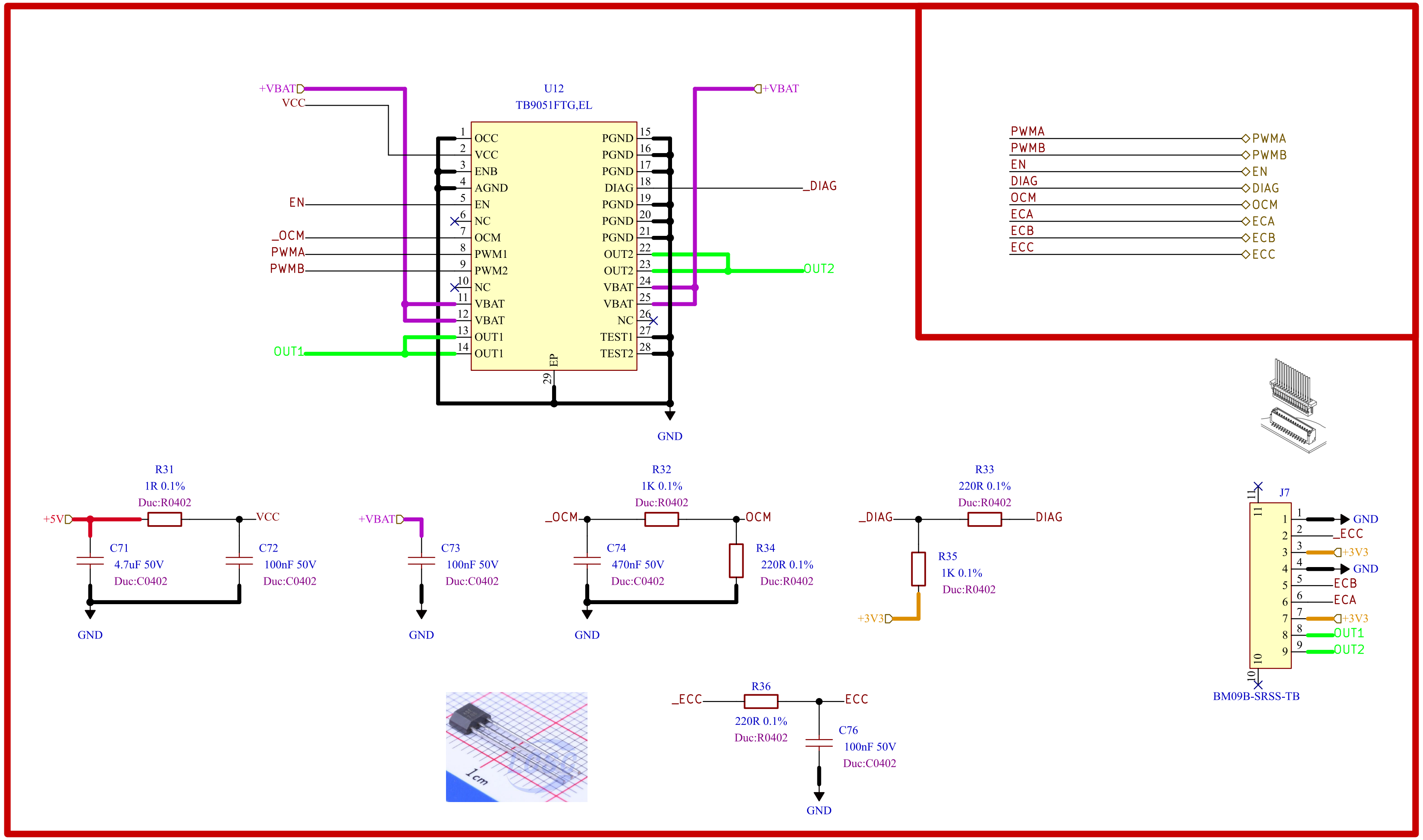
Comments:	Company: Willas 威倫電子股份有限公司 Willas Electronic Corp.			Variant: DRAFT	Git Hash: 309c1de
	Board Name: **Motor Control Board**			Project Name: **Master Hand**	
Sheet Title:	File Name: Motor Control.kicad_sch	Designer: Damon		Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Motor Control 3/		Reviewer:		Size: A4	Sheet: 10 of 23

[11] Motor Control 4



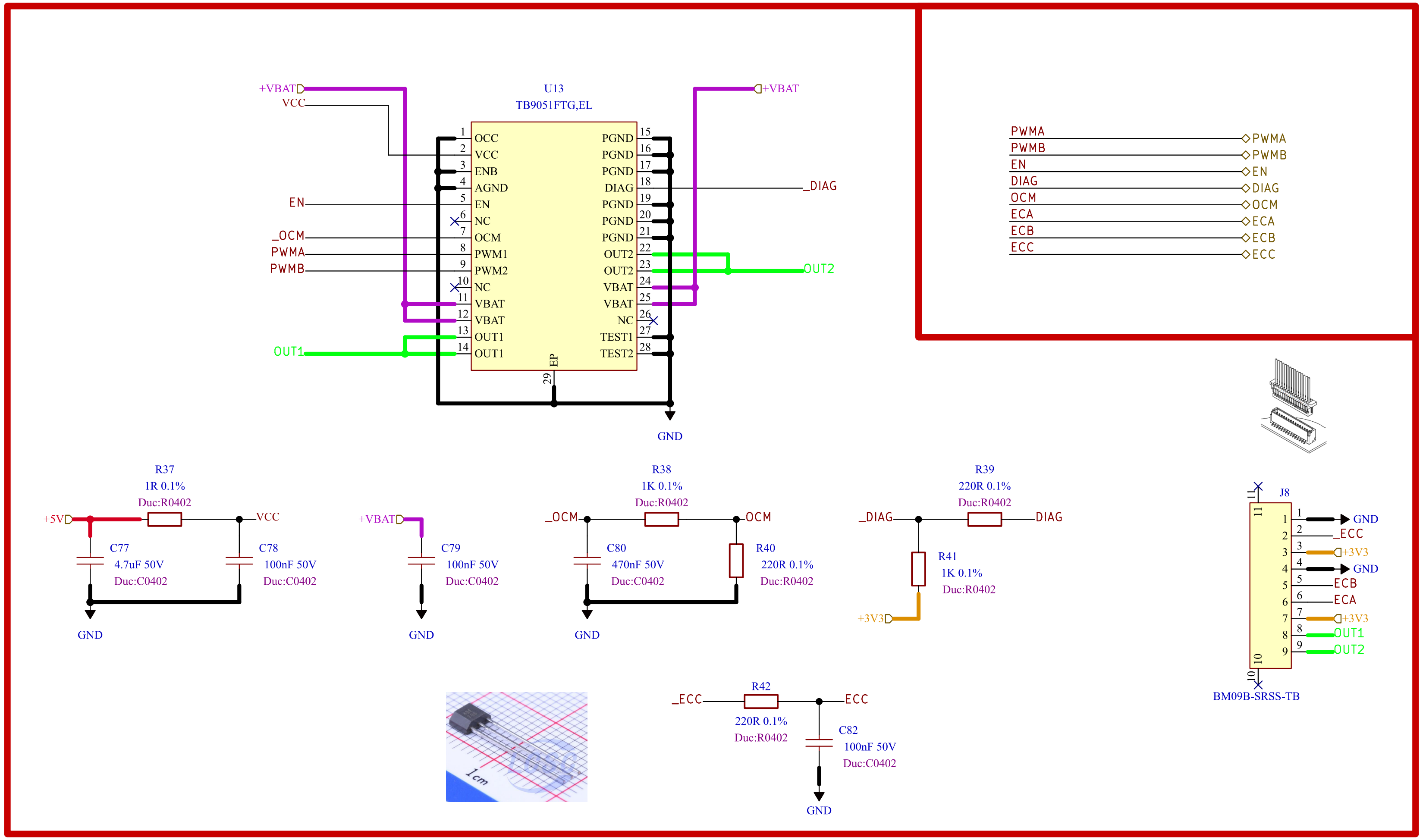
Comments:	Company: Willas — 威倫電子股份有限公司 Willas Electronic Corp.		Variant: DRAFT	Git Hash: 309c1de
	Board Name: **Motor Control Board**		Project Name: **Master Hand**	
Sheet Title:	File Name: Motor Control.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Motor Control 4/		Reviewer:	Size: A4	Sheet: 11 of 23

[12] Motor Control 5



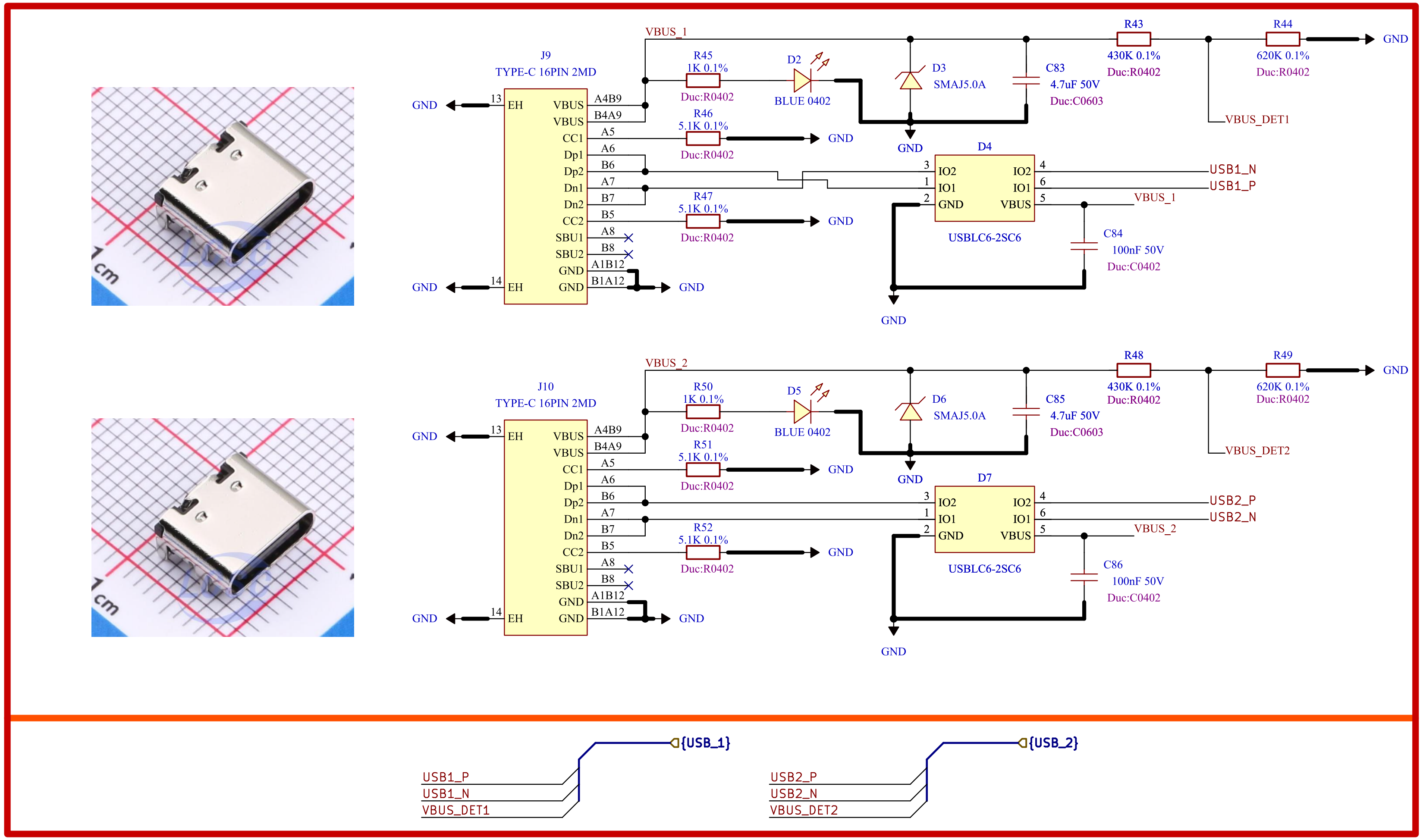
Comments:	Company: Willas	威倫電子股份有限公司 Willas Electronic Corp.		Variant: DRAFT	Git Hash: 309c1de
	Board Name: **Motor Control Board**			Project Name: **Master Hand**	
Sheet Title:	File Name: Motor Control.kicad_sch		Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Motor Control 5/			Reviewer:	Size: A4	Sheet: 12 of 23

[13] Motor Control 6



Comments:	Company: Willas (威倫電子股份有限公司 Willas Electronic Corp.)		Variant: DRAFT	Git Hash: 309c1de
	Board Name: **Motor Control Board**		Project Name: **Master Hand**	
Sheet Title:	File Name: Motor Control.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Motor Control 6/		Reviewer:	Size: A4	Sheet: 13 of 23

[14] Interface - USB 2.0



DESIGN NOTE:
[1] VBUS NOT connected to +5V/+3V3. Board is self-powered(LiPo). Sense + LED only. No backfeed to host.

[2] LED direct from VBUS @ 1mA. USB2.0 suspend limit 2.5mA. LED 1mA + divider 33uA = 1.03mA (41%). Do NOT raise to 5mA.

[3] VBUS_DET required - G474 has no dedicated VBUS pin. Self-powered device must not pull up D+ when VBUS=0. FW controls DPPU bit in USB_BCDR. 68k/100k: VBUS 4.40V -> 2.60V > VIH 2.31V. Debounce 100ms.

[4] D+/D- direct to PA12/PA11. No 22R series. No external 1k5 pull-up (internal RPUI 900-1500R).

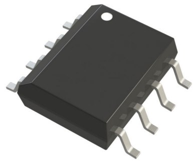
[5] USB clock: PLL-Q 48MHz from HSE 8MHz. ST DFU bootloader uses HSI48+CRS - flashable without crystal.

[6] Both USB ports fully isolated - no shared nets.

[7] GROUND LOOP WARNING: USB from PC ties host GND to power GND (4.6A peak @20kHz). Use isolator (ADuM3160) or isolated USB-UART when debugging with motors running.

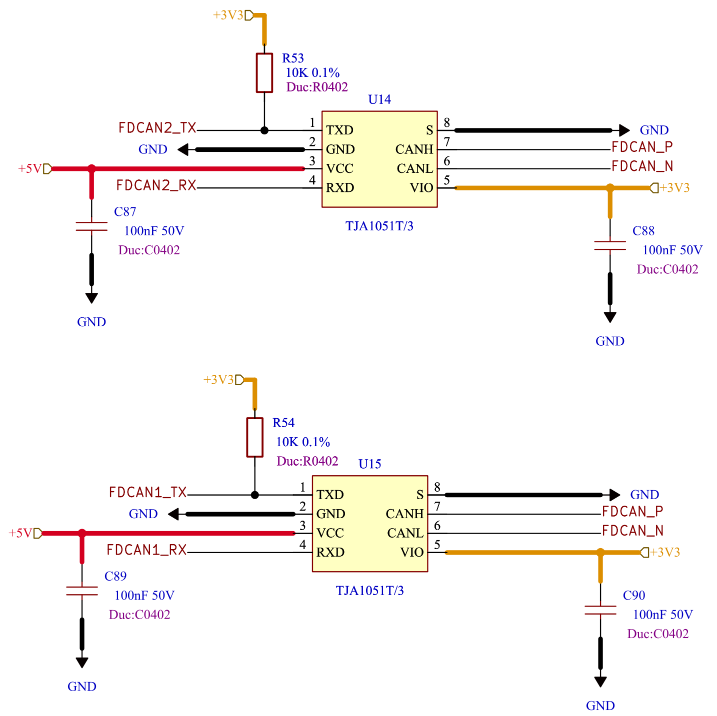
Comments:	Company: Willas (威倫電子股份有限公司 Willas Electronic Corp.)		Variant: DRAFT	Git Hash: 309c1de
	Board Name: **Motor Control Board**		Project Name: **Master Hand**	
Sheet Title:	File Name: Interface - USB 2.0.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Interface - USB 2.0/		Reviewer:	Size: **A4**	Sheet: **14** of **23**

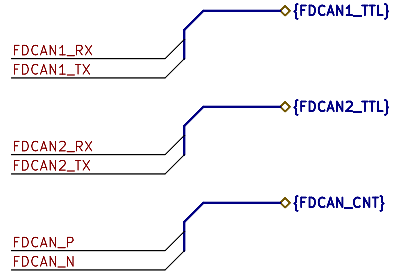
[15] Interface - FD-CAN



DESIGN NOTE:
Silent mode disabled. Pin "S" to ground.

DESIGN NOTE:
120-ohm termination in Interconnects sheet.





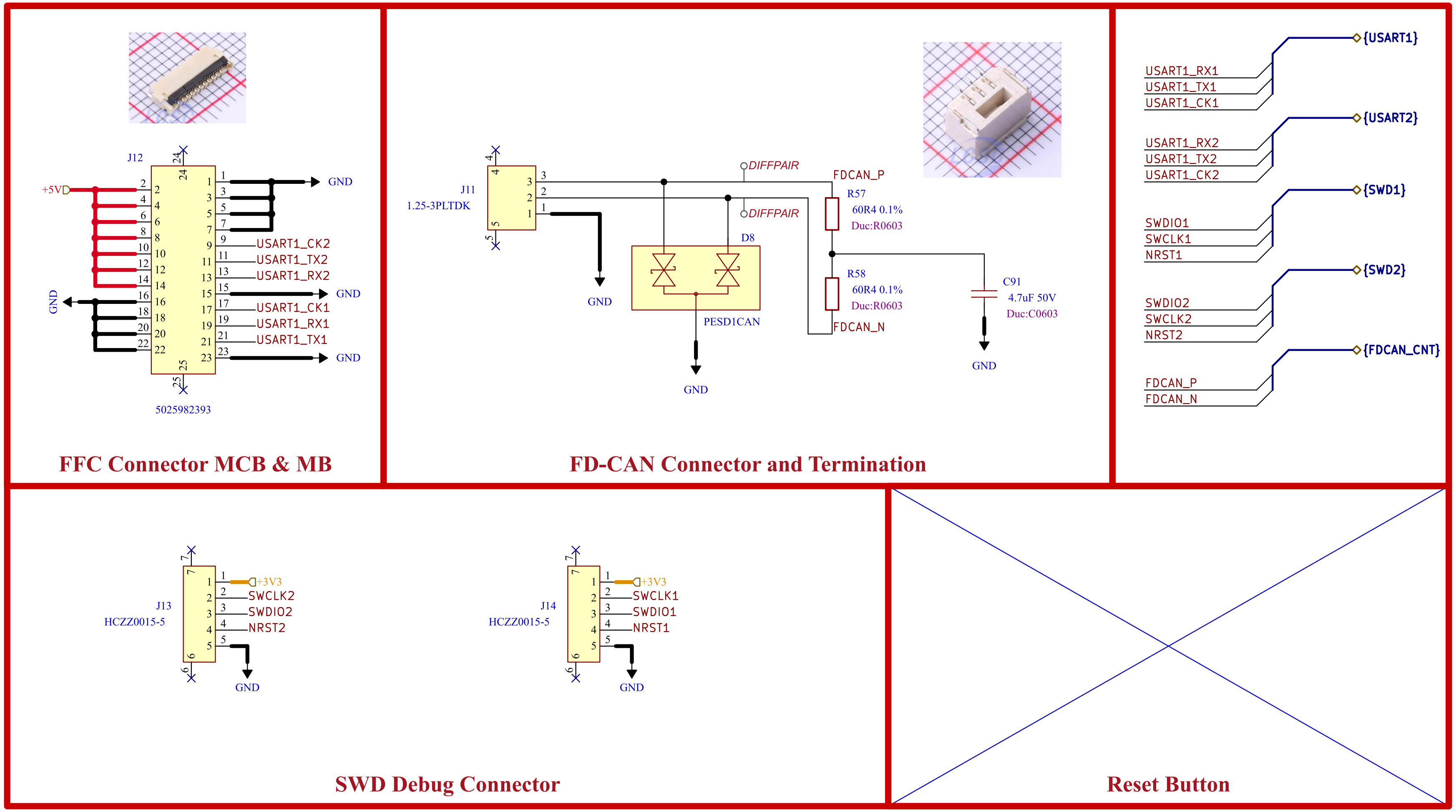
LAYOUT NOTE:
Place close to FD-CAN connector if possible to reduce stub length.

LAYOUT NOTE:
We decided to route it further away to minimize risk of ground lift on the RX/TX lines due to high currents.

Fault Protected FD-CAN Transceiver

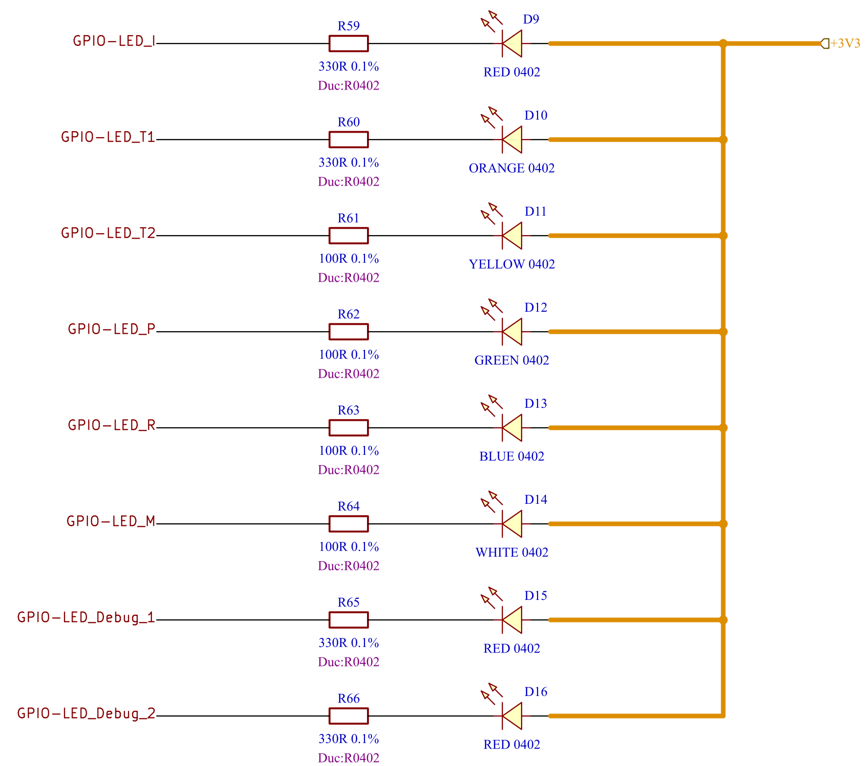
Comments:	Company: Willas (威倫電子股份有限公司 Willas Electronic Corp.)		Variant: DRAFT	Git Hash: 309c1de
	Board Name: Motor Control Board		Project Name: Master Hand	
Sheet Title:	File Name: Interface - FD-CAN.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Interface - FD-CAN/		Reviewer:	Size: A4	Sheet: 15 of 23

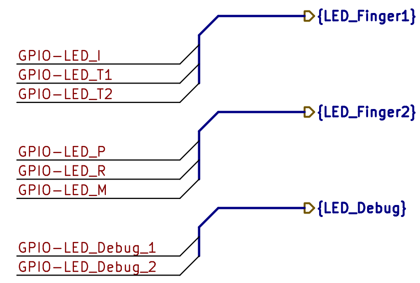
[16] Interface - Interconnects



Comments:	Company: Willas	威倫電子股份有限公司 Willas Electronic Corp.	Variant: DRAFT	Git Hash: 309c1de
	Board Name: **Motor Control Board**		Project Name: **Master Hand**	
Sheet Title:	File Name: Interface - Interconnects.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Interface - Interconnects/		Reviewer:	Size: A4	Sheet: 16 of 23

[17] User - LED Indicators & Buzzer

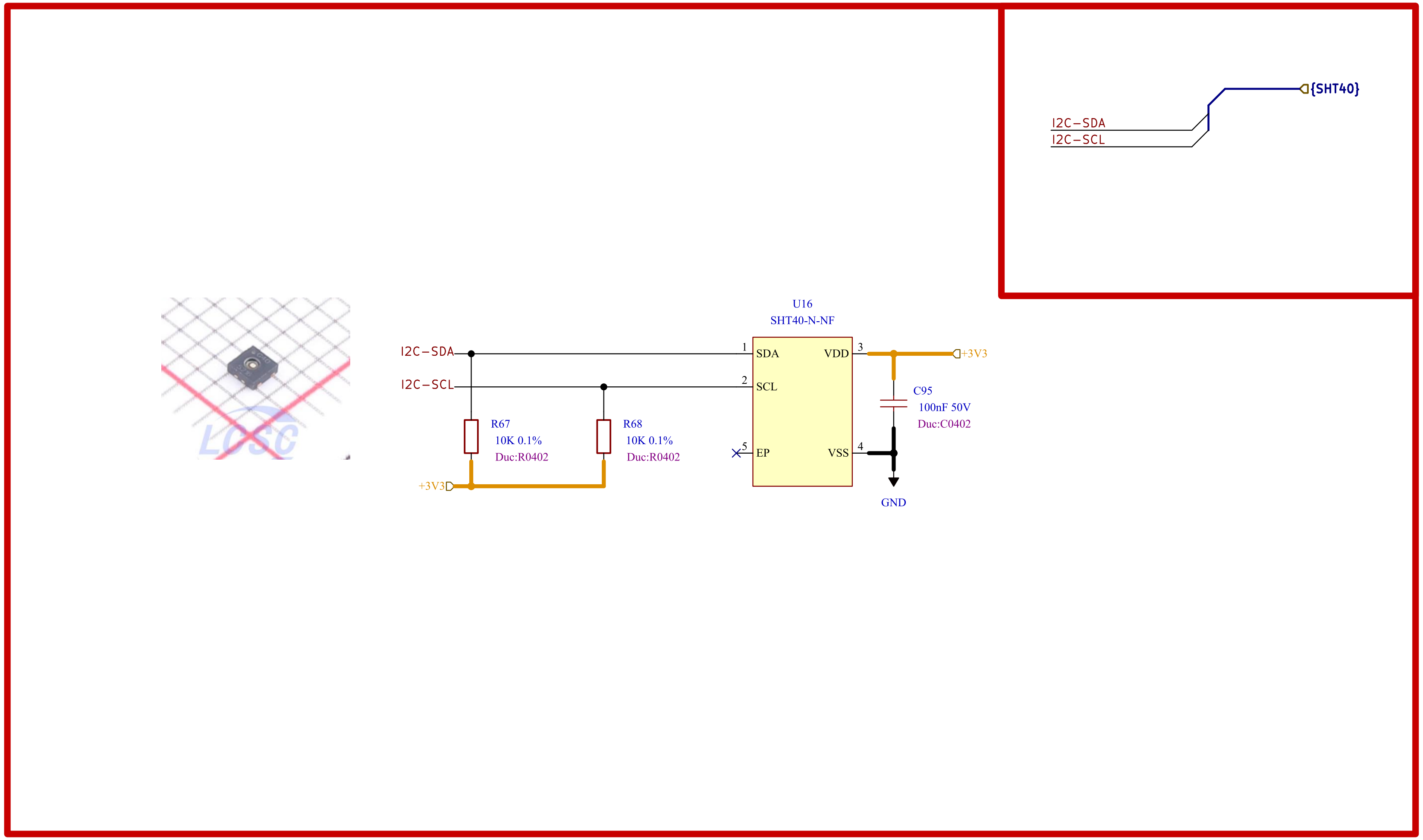




LED Color	Typical Vf Range
Red	1.8 to 2.1
Amber	2 to 2.2
Orange	1.9 to 2.2
Yellow	1.9 to 2.2
Green	2 to 3.1
Blue	3 to 3.7
White	3 to 3.4

Comments:	Company: Willas (威倫電子股份有限公司 Willas Electronic Corp.)			Variant: DRAFT	Git Hash: 309c1de
	Board Name: **Motor Control Board**			Project Name: **Master Hand**	
Sheet Title:	File Name: User - LED Indicators & Buzzer.kicad_sch		Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/User - LED Indicators & Buzzer/			Reviewer:	Size: **A4**	Sheet: **17** of **23**

[18] Sensing - Temperature & Humidity



Comments:	Company: Willas		Variant: DRAFT	Git Hash: 309c1de
	Board Name: **Motor Control Board**		Project Name: **Master Hand**	
Sheet Title:	File Name: Sensing - Temperature.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Sensing - Temperature & Humidity/		Reviewer:	Size: A4	Sheet: 18 of 23

[19] Sensing - Battery

DESIGN NOTE:
Battery voltage sensing divider for 2S LiPo battery.

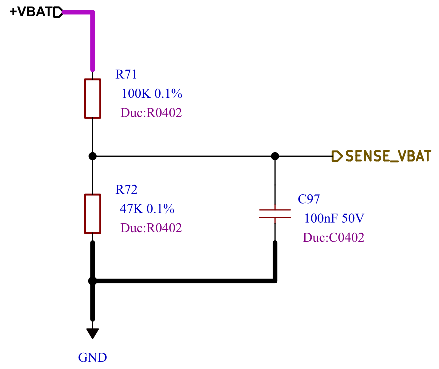
R1 = 100kΩ (Top)
R2 = 47kΩ (Bottom)

Divider Ratio:
VADC = VBAT × (47 / (100 + 47))
VADC = VBAT × 0.3197

2S LiPo:
Max battery voltage (8.4V): 2.69 V
Nominal voltage (7.4V): 2.37 V
Low battery voltage (6.4V): 2.05 V
Cutoff voltage (6.0V): 1.92 V

Maximum divider current:
I = 8.4V / 147kΩ = 57 µA

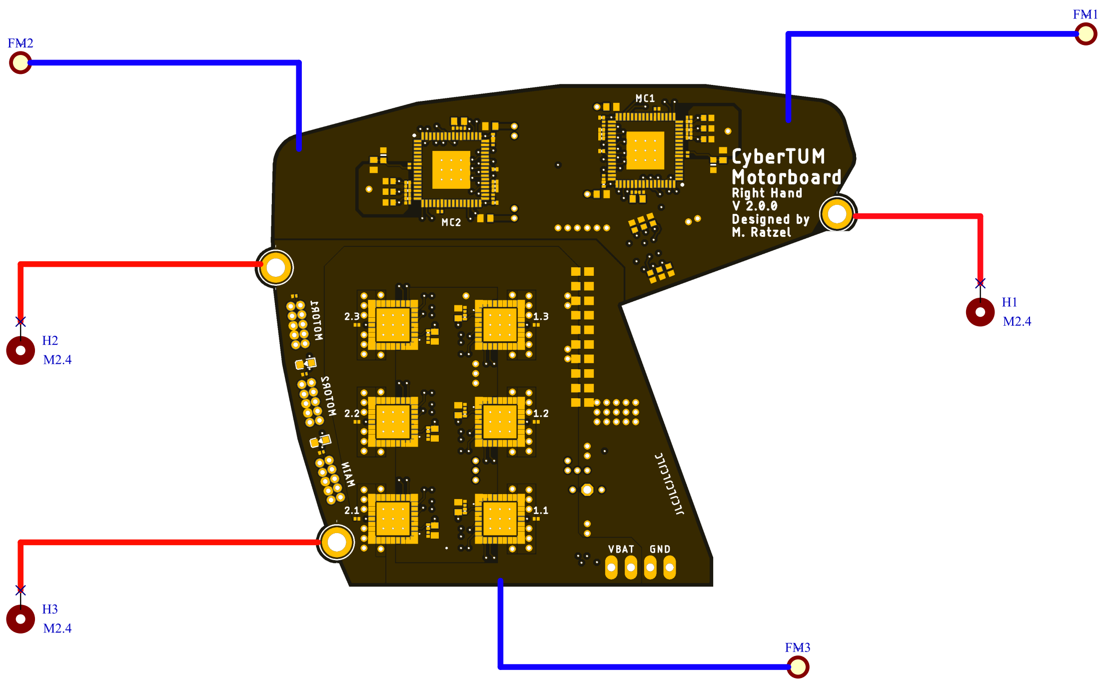
Notes:
- Designed for 3.3V ADC input.
- Add 100nF capacitor at ADC input for noise filtering.
- Use 1% resistors or better.
- Use low temperature coefficient resistors (≤100 ppm/°C) for improved measurement accuracy.



Battery Voltage Sensing

Comments:	Company: Willas (威倫電子股份有限公司 Willas Electronic Corp.)		Variant: DRAFT	Git Hash: 309c1de
	Board Name: **Motor Control Board**		Project Name: **Master Hand**	
Sheet Title:	File Name: Sensing - Battery.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Sensing - Battery/		Reviewer:	Size: **A4**	Sheet: **19** of **23**

[20] Misc - Holes, Fiducials



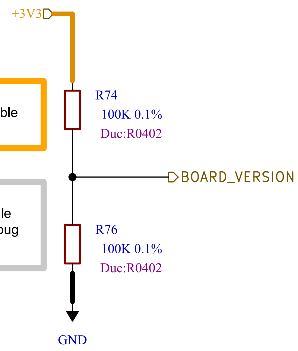
Mounting Holes and Fiducial Markers

Comments:	Company: Willas		Variant: DRAFT	Git Hash: 309c1de	
	Board Name: **Motor Control Board**		Project Name: **Master Hand**		
Sheet Title:	File Name: Misc - Holes Fiducials.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)	
Sheet Path: /Project Architecture/Misc - Holes, Fiducials/		Reviewer:	Size: **A4**	Sheet: **20** of **23**	

[21] Misc - Board Version, Switch

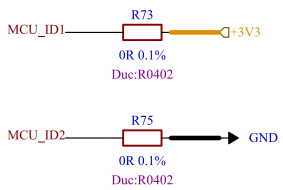
DESIGN NOTE:
Necessary to be compatible with BH FW.

DESIGN NOTE:
Pin is also used at compile time to output analog debug data (DAC).

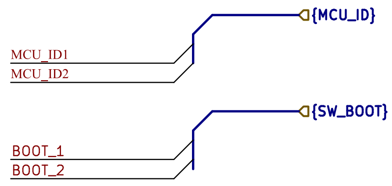


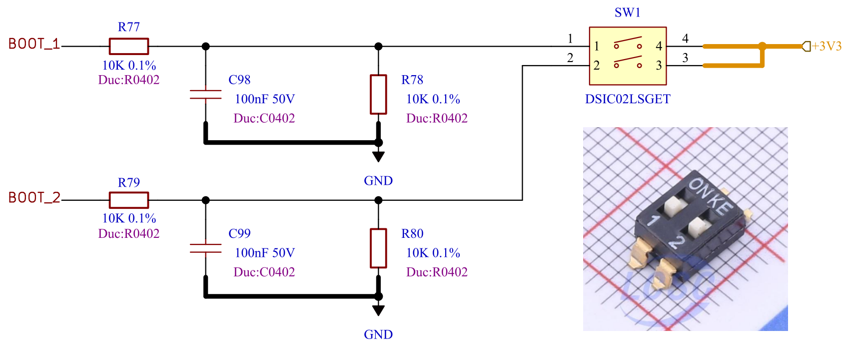
		TOP	BOTTOM
MCB	r1.1, r1.2	×	100k
	r1.3	100k	×
MB	v1.0-1.1	100k	100k

Board Version Resistor Divider



MCU ID



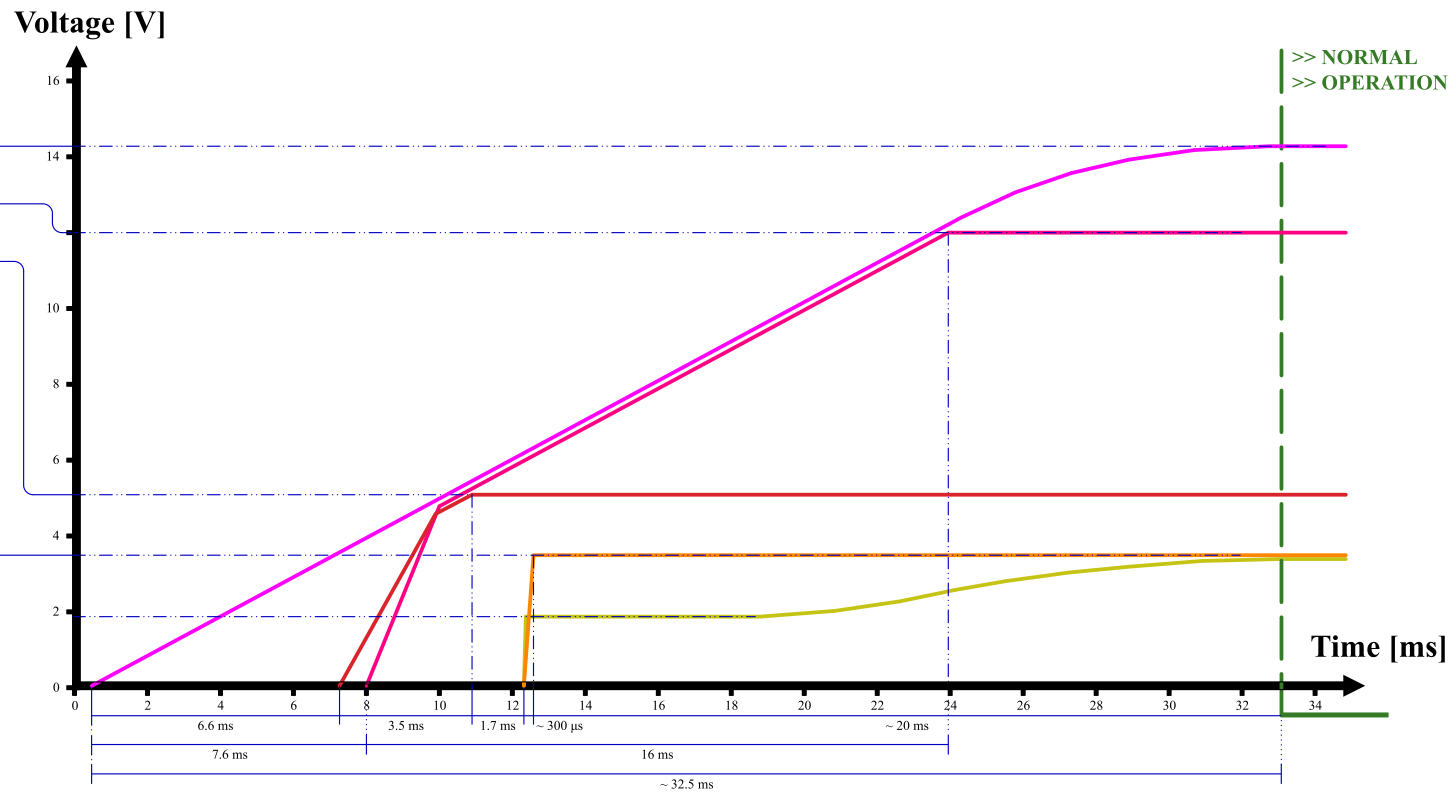


SWITCH BOOT MODE

Comments:	Company: Willas (威倫電子股份有限公司 Willas Electronic Corp.)	Variant: DRAFT	Git Hash: 309c1de
	Board Name: Motor Control Board	Project Name: Master Hand	
Sheet Title:	File Name: Misc - Board Version, Switch.kicad_sch	Designer: Damon	Date: 2026-06-27 / Revision: + (Unreleased)
Sheet Path: /Project Architecture/Misc - Board Version, Switch/		Reviewer:	Size: A4 / Sheet: 21 of 23

[22]

NAME	SOURCE	LEVEL
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%



Comments:	Company: Willas	Variant: DRAFT	Git Hash: 309c1de
	Board Name: **Motor Control Board**	Project Name: **Master Hand**	
Sheet Title:	File Name: Power - Sequencing.kicad_sch	Designer: Damon	Date: 2026-06-27 / Revision: + (Unreleased)
Sheet Path: /Power - Sequencing/		Reviewer:	Size: A4 / Sheet: 22 of 23

[23]

Comments:	Company: Willas — 威倫電子股份有限公司 Willas Electronic Corp.		Variant: DRAFT		Git Hash: 309c1de
	Board Name: **Motor Control Board**		Project Name: **Master Hand**		
Sheet Title:	File Name: Revision History.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)	
Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 23 of 23	